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EXP4: Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.
index.html

```
<!DOCTYPE html>

<html lang="en">


<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Experiment 4</title>

  <!-- Bootstrap CSS -->

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet"

    integrity="sha384-
QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhjY6hW+ALEwIH"
crossorigin="anonymous">

  <!-- Local CSS -->

  <link rel="stylesheet" href="style.css">

  <!-- jQuery CDN -->

  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

</head>


<body class="bg-light">


  <div class="container mt-5">

    <header class="text-center mb-5">

      <h1>Experiment 4: jQuery Effects</h1>

      <p class="text-muted">Dynamic DOM & jQuery Animation</p>

    </header>


    <main>
```

```
<div class="row justify-content-center">
  <div class="col-md-8">
    <!-- Display Area -->
    <div class="card mb-4 shadow-sm" id="target-box">
      <div class="card-body text-center p-5">
        <h2 id="dynamic-title" class="card-title">Hello, World!</h2>
        <p id="dynamic-text" class="card-text">This is a dynamic element ready for interaction.</p>
      </div>
    </div>

    <!-- Controls -->
    <div class="card">
      <div class="card-body">
        <div class="row g-3 text-center">
          <div class="col-md-4">
            <h5>DOM Manipulation</h5>
            <button id="btn-update" class="btn btn-primary w-100">Update Content</button>
          </div>

          <div class="col-md-4">
            <h5>jQuery Visibility</h5>
            <div class="btn-group w-100" role="group">
              <button id="btn-hide" class="btn btn-danger">Hide</button>
              <button id="btn-show" class="btn btn-success">Show</button>
            </div>
          </div>

          <div class="col-md-4">
            <h5>jQuery Animation</h5>
            <button id="btn-animate" class="btn btn-info w-100 text-white">Animate</button>
          </div>
```

```
            </div>
          </div>
        </div>
      </div>
    </div>
  </main>
 </div>

 <!-- Bootstrap Bundle JS -->
 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
  integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDzOxhy9GkcIdslK1eN7N6jIeHz"
  crossorigin="anonymous"></script>
 <script src="script.js"></script>
</body>

</html>
```

style.css

```
/* Basic styling overriding Bootstrap defaults for a simple look */
body {
   font-family: Arial, sans-serif;
   padding-top: 50px;
   background-color: #f8f9fa; /* Consistent light background */
}

/* Adjust the main card to look plain */
#target-box {
   border: 1px solid #dee2e6;
   background-color: white;
   border-radius: 5px;
```

```
}

/* Make header simpler */
header h1 {
    font-size: 2rem;
    color: #333;
}

/* Ensure controls are spaced out but not fancy */
.card-body {
    padding: 1.5rem;
}

button {
    margin-top: 10px;
}
```

script.js

```
// DOM Content Loaded - ensures HTML and scripts are fully parsed before running
document.addEventListener('DOMContentLoaded', () => {

    // --- 1. DOM Manipulation (Vanilla JS) ---
    // Objective: Update content dynamically on button click

    const updateBtn = document.getElementById('btn-update');
    const titleElement = document.getElementById('dynamic-title');
    const textElement = document.getElementById('dynamic-text');
    const displayBox = document.getElementById('target-box'); // Changed from glass-card

    const contentVariants = [
```

```
        { title: "Updated!", text: "Content changed via JavaScript." },
        { title: "Another Click", text: "Here is some new text." },
        { title: "DOM Demo", text: "Basic DOM manipulation in action." },
        { title: "Experiment 4", text: "Click to cycle through text." }
    ];

    let contentIndex = 0;

    updateBtn.addEventListener('click', () => {
        // Cycle through content
        contentIndex = (contentIndex + 1) % contentVariants.length;
        const newContent = contentVariants[contentIndex];

        // Update the DOM elements
        titleElement.textContent = newContent.title;
        textElement.textContent = newContent.text;

        // Simple highlight effect (gold background flash)
        displayBox.style.backgroundColor = "#fff3cd";
        setTimeout(() => {
            displayBox.style.backgroundColor = "white";
        }, 300);
    });
});

// --- 2. jQuery Effects ---
// Objective: Implement Hide, Show, Toggle, and Animate
$(document).ready(function() {

    const targetBox = $('#target-box');
```

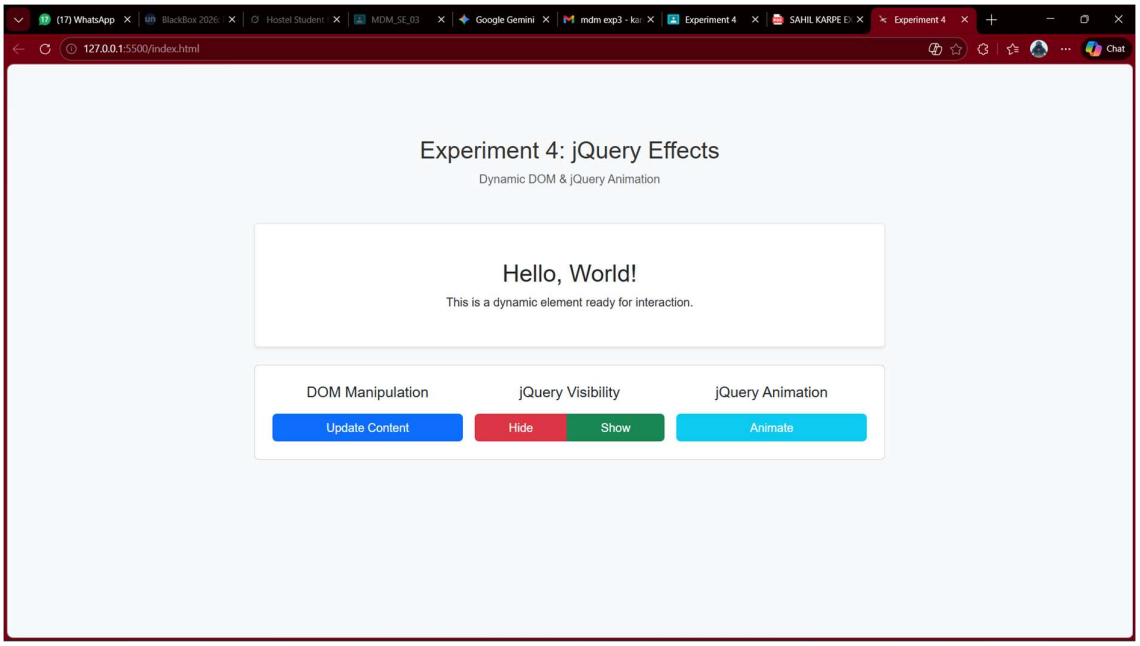
```
    // Hide Effect
    $('#btn-hide').click(function() {
        targetBox.hide(); // Default fast hide
    });

    // Show Effect
    $('#btn-show').click(function() {
        targetBox.show(); // Default fast show
    });

    // Animate Effect
    $('#btn-animate').click(function() {
        const box = targetBox;

        // Simple, standard jQuery animation
        box.animate({
            opacity: 0.5,
            marginLeft: "50px",
            height: "toggle" // classic jQuery animation trick
        }, 1000, function() {
            // Callback to reset or do something else
            box.animate({
                opacity: 1,
                marginLeft: "0px",
                height: "toggle"
            }, 1000);
        });
    });

});
```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BlackBox 2026
Hostel Student
MDM_SE_03
Google Gemini
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Experiment 4
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Experiment 4
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Chat
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jQuery Visibility
jQuery Animation
Update Content
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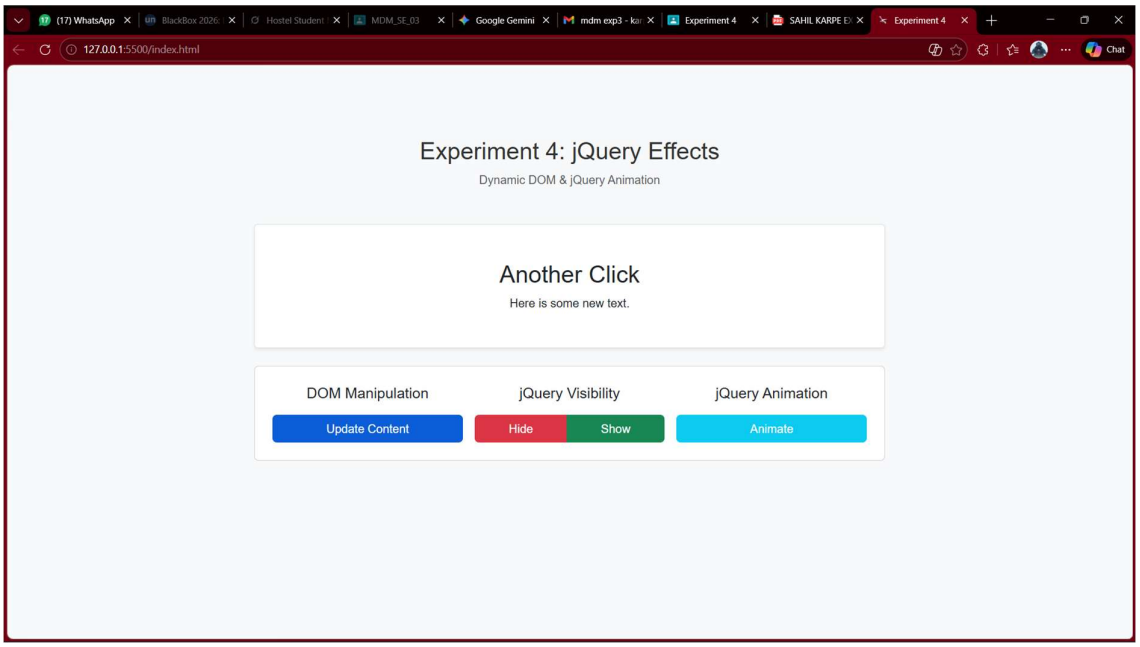
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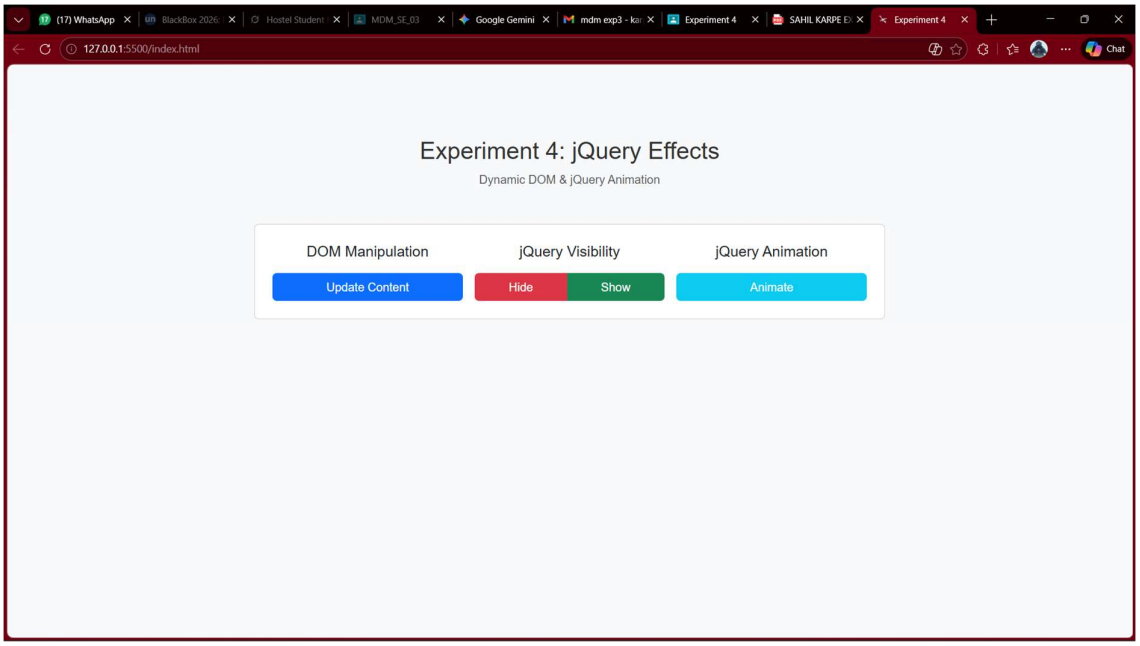
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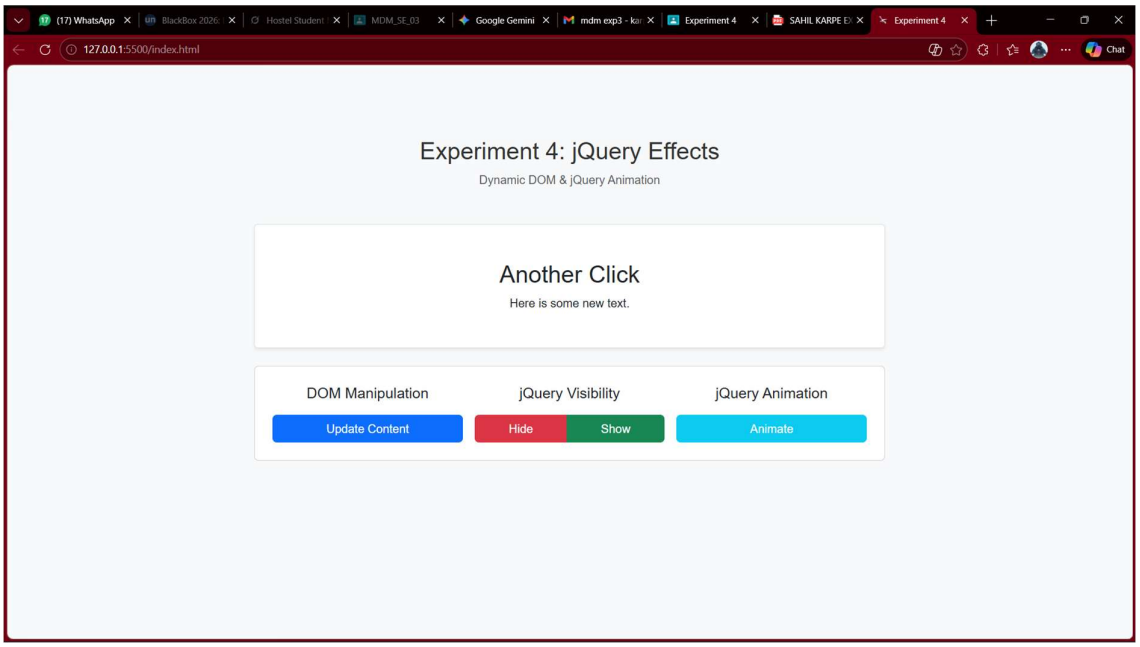
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